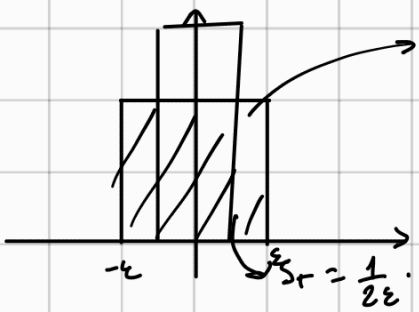
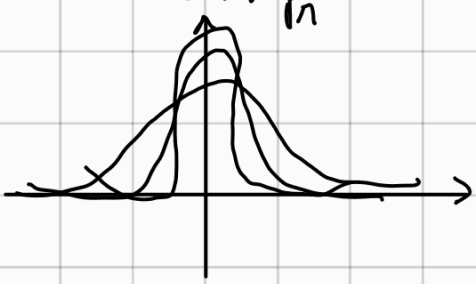
$$\delta(x-a) = \begin{cases} x \neq a, 0 \\ x = a \quad \infty \end{cases}$$



$$f_\varepsilon(x) = \begin{cases} 0\,; |x| > \varepsilon \\ 1/2\varepsilon \quad |x| < \varepsilon \end{cases} \qquad \int_{-\infty}^{\infty} \delta(x-a)\,dx = 1$$

$$S_\tau = \frac{1}{2\varepsilon} \cdot 2\varepsilon = 1 \Rightarrow \lim_{\varepsilon \to 0} S_\tau = \lim_{\varepsilon \to 0} \frac{1}{2\varepsilon} 2\varepsilon = \lim_{\varepsilon \to 0} 1 = 1$$

$$\delta(x) = \lim_{z \to a} \frac{z}{\sqrt{\pi}} e^{-z^2 x^2} = \begin{cases} 0\,, x \neq 0 \\ \infty\,, x = 0 \end{cases}$$



$$\int_{-\infty}^{\infty} \delta(x)\,dx = 1 = \int_{-\infty}^{\infty} \lim_{z \to \infty} \frac{z}{\sqrt{\pi}} e^{-z^2 x^2} dx = \frac{1}{\sqrt{\pi}} \lim_{z \to \infty} z \int_{-\infty}^{\infty} e^{-z^2 x^2} dx$$

$$zx = x' \Rightarrow dx = \frac{dx'}{z}$$

$$\int_{-\infty}^{\infty} \delta_z(x)\,dx = \int_{-\infty}^{\infty} \lim \frac{z}{\sqrt{\pi}} e^{-z^2 x^2} dx$$

$$= \frac{1}{\sqrt{\pi}} \lim_{z \to \infty} z \int_{x=-\infty}^{\infty} e^{-x^2} \frac{dx'}{z}$$

$$= \frac{1}{\sqrt{\pi}} \lim_{z \to \infty} z \cdot \frac{1}{z} \int_{-\infty}^{\infty} e^{-x'^2}$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x'} dx' = \frac{\sqrt{\pi}}{\sqrt{\pi}} = 1$$

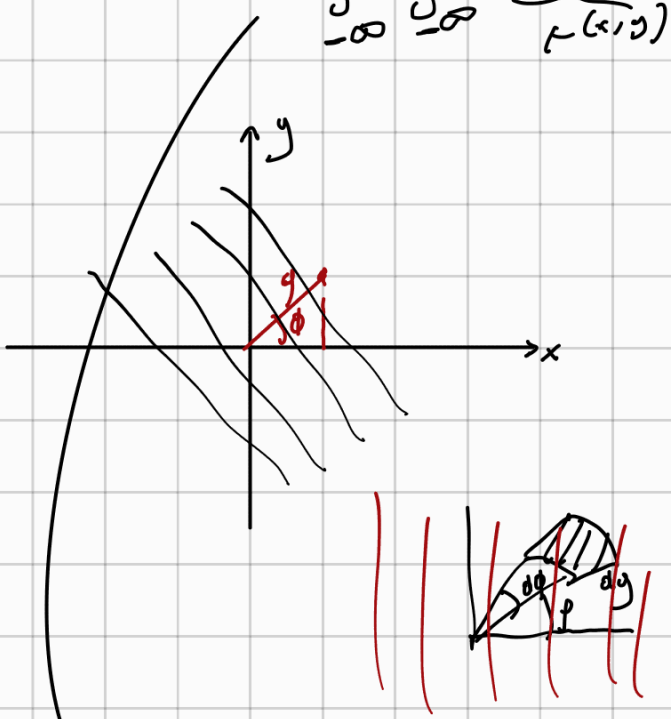


$$I = \int_{-\infty}^{\infty} e^{-x^2} dx = \int_{-\infty}^{\infty} e^{-y^2} dy \quad > 0$$

$$\int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y} dy = I^2$$

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2-y^2} dx\,dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \underbrace{e^{-(x^2+y^2)}}_{f(x,y)} dx\,dy$$



$$e^{-(x^2+y^2)} = e^{-\rho^2}$$

$$ds = dx\,dy$$

$$ds = \rho\, d\phi\, d\rho$$

$$0 < \rho < \infty$$

$$0 < \phi < 2\pi$$

$$\int_{\rho=0}^{\infty} \int_{\phi=0}^{2\pi} e^{-\rho^2} \rho\, d\rho\, d\phi = \frac{2\pi}{-2} \int_{\rho=0}^{\infty} -2\rho e^{-\rho^2} d\rho$$

$$I^2 = -\pi \int_{\rho=0}^{\infty} -2\rho e^{-\rho^2} d\rho = -\pi e^{-\rho^2} \Big|_{\rho=0}^{\infty} = -\pi (e^{-\infty} - e^{0}) = \pi$$

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx = 2\int_{0}^{\infty} e^{-x^2} dx = \sqrt{\pi} \rightarrow \int_{0}^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

* $f(x)\,\delta(x-a) = f(a)\,\delta(x-a)$

* $\int_{-\infty}^{\infty} f(x)\,\delta(x-a)\,dx = \int_{-\infty}^{\infty} f(a)\,\delta(x-a)\,dx$

$= f(a)\int_{-\infty}^{\infty}\delta(x-a)\,dx = f(a)$

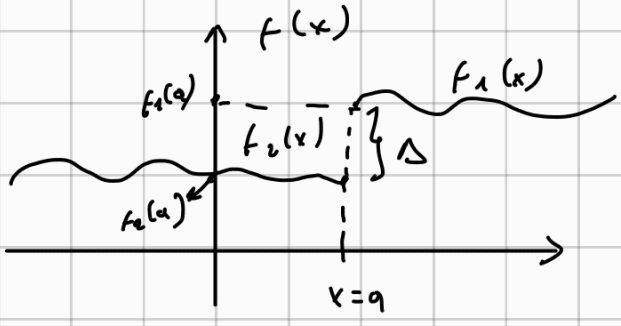
$\int_{-\infty}^{\infty}\delta(x-a)\,dx = 1 = \int_{\xi_1}^{\xi_2}\delta(x-a)\,dx = \begin{cases} 1, & \xi_1 < a < \xi_2 \\ 0, & \text{aksi halde} \end{cases}$

$f(x) = \int_{-\infty}^{x}\delta(z-a)\,dz = \begin{cases} 1; & a < x \\ 0; & a > x \end{cases} = u(x)$



$u(x-a) \triangleq \int_{-\infty}^{x}\delta(z-a)\,dz \Rightarrow \frac{du(x-a)}{d(x-a)} = u'(x-a) = \delta(x-a)$

$u'(x) \triangleq \delta(x)$



$f(x) = \begin{cases} f_1(x): x > a \\ f_2(x): x < a \end{cases} = f_1(x)\,u(x-a) + f_2(x)\,[1-u(x-a)]$

$f_1(a) - f_2(a) = \Delta$ $\quad f'(x) = \frac{df(x)}{dx} = \frac{d}{dx}[f_1(x)\,u(x-a)] + \frac{d}{dx}[f_2(x)[1-u(x-a)]]$

$= f_1'(x)\,u(x-a) + f_1(x)\,u'(x-a) + f_2'(x)[1-u(x-a)] + f_2(x)[1-u(x-a)]'$

$= f_1'(x)\,u(x-a) + f_2'(x)[1-u(x-a)] + f_1(x)\,\delta(x-a) - f_2(x)\,\delta(x-a)$

$= \begin{cases} f_1'(x); & x > a \\ f_2'(x); & x < a \end{cases} + [f_1(x) - f_2(x)]\,\delta(x-a)$

$= \begin{cases} f_2'(x); & x > a \\ f_2'(x); & x < a \end{cases} + [f_1(a) - f_2(a)]\,\delta(x-a)$

tekil kısmı distribüsyondur

$$V = \begin{cases} V_1(x,y,z); & x > a \\ V_2(x,y,z) & x < a \end{cases} \Rightarrow \operatorname{grad} V = \begin{cases} \operatorname{grad} V_1; & x>0 \\ \operatorname{grad} V_2; & x<a \end{cases} + \vec{e}_x \underbrace{[V_1 - V_2]\big|_{x=a}}_{\Delta V|_{x=a}} \delta(x-a)$$

$$= \vec{e}_x (V_1 - V_2)|_{x=a}$$

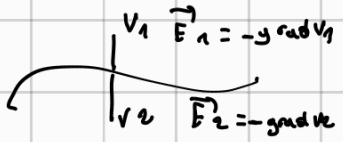
$$= -\vec{e}_x (V_2 - V_1)|_{x=a}$$

$$\vec{D} = \begin{cases} \vec{D}_1, & x > a \\ \vec{D}_2, & x < a \end{cases}$$

$$\operatorname{div} \vec{D} = \begin{cases} \operatorname{div} \vec{D}_1; & x > a \\ \operatorname{div} \vec{D}_2; & x < a \end{cases} + \vec{e}_x (\vec{D}_1 - \vec{D}_2)\big|_{x=a} \delta(x-a)$$

$$\vec{n} \Delta f|_{S'} \delta(S')$$

$$\vec{E} = \begin{cases} \vec{E}_1, & x > a \\ \vec{E}_2, & x < a \end{cases} \Rightarrow \operatorname{rot} \vec{E} = \begin{cases} \operatorname{rot} \vec{E}_1; & x > a \\ \operatorname{rot} \vec{E}_2; & x < a \end{cases}$$



$$\vec{E} = -\operatorname{grad} V = \begin{cases} \vec{E}_1 \\ \vec{E}_2 \end{cases} + \vec{n}[V_1 - V_2]|_S \delta(S)$$

$$\downarrow \vec{n}[V_1|_S - V_2|_S] \delta(G)$$

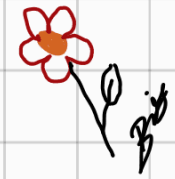
$$\vec{D}_1 = \varepsilon_1 \vec{E}_1$$

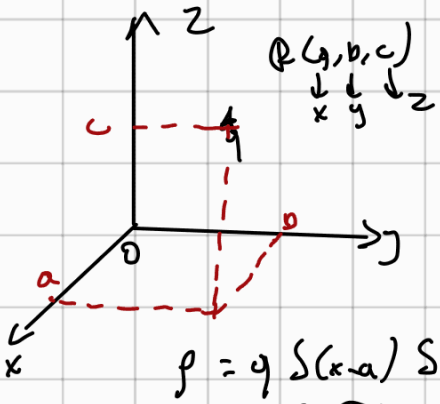
$$\vec{D}_2 = \varepsilon_2 \vec{E}_2$$

$$\vec{D} = \begin{cases} \vec{D}_1; & \vec{\omega} \text{ links } (S) \\ \vec{D}_2; & \text{alt } (S) \end{cases}$$

$$\operatorname{div} \vec{D} = \begin{cases} \operatorname{div} \vec{D}_1 & \vec{\omega} \text{ links } (S) \\ \operatorname{div} \vec{D}_2 & \text{alt } (S) \end{cases} + \underbrace{\vec{n}(\vec{D}_1 - \vec{D}_2)\big|_{S'}}_{\rho_S} \delta(S)$$

$$\rho = \begin{Bmatrix} \rho_1 \\ \rho_2 \end{Bmatrix} + \rho_S \delta(s)$$

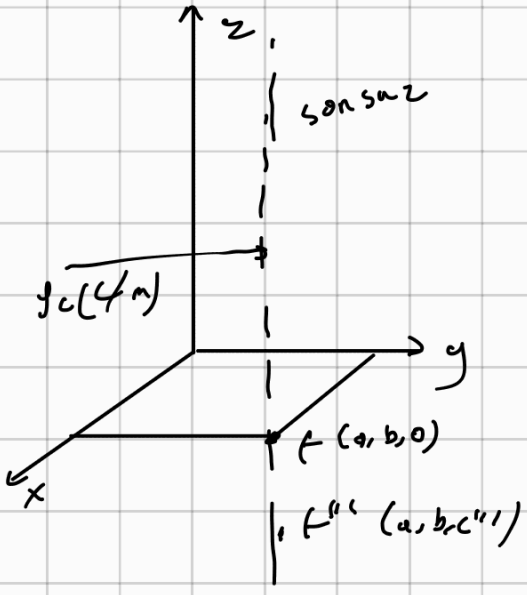




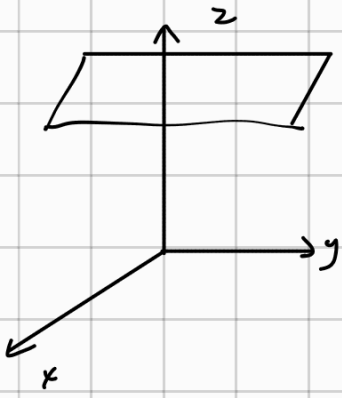
$$\rho = q\,\delta(x-a)\,\delta(y-b)\,\delta(z-c) \quad (C/m^3)$$

$\delta(x-a)$: $1/m$, $\delta(y-b)$: $1/m$, $\delta(z-c)$: $1/m$

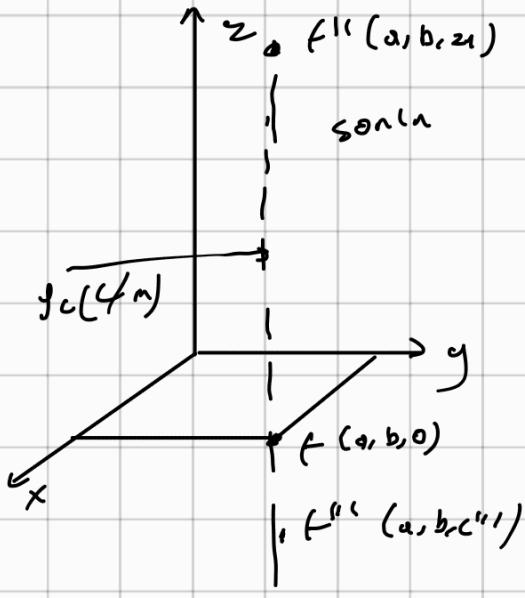
$\rho = \frac{dq}{dv} = \infty$



$$\rho = \rho_S\,\delta(x-a)\,\delta(y-b)$$



$$\rho = \rho_S\,\delta(z-a) \quad (C/m^3)$$

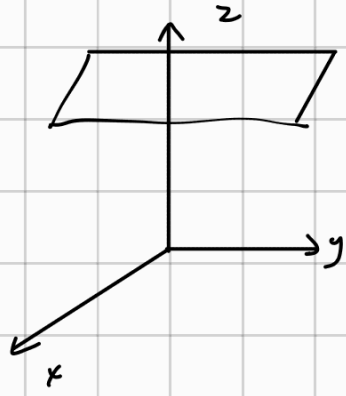


$$\rho = \rho_c\, S(x-a)\, S(y-b)\, [u(z-z_1) + u(z-z_2)]$$
$$= \rho_c\, S(x-a)\, S(y-b) \qquad z_2 < z < z_1$$

$$\sin x\, S(x) = \sin 0\, S(x) = 0$$

$$\cos x\, S(x) = \cos S(x) = S(x)$$

$$\tan(x)\ S(x-\pi)$$
$$= \tan(\pi) S(x-\pi) = 0 \ //$$



$$\rho = \rho_s\, S(z-a)\, [u(y-y_1) - u(y-y_2)]\, [u(x-x_2) - u(x-x_1)]$$
$$y < y_1 < y_2$$